Random Vectors and Samples

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Content adapted from:
Johnson, R. A., & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th ed).

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Definition of Random Vectors and Matrices

A random vector is a vector whose elements are random variables:

$$\boldsymbol{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$$

where X_j is a random variable with some distribution F_j.

A random matrix is a matrix whose elements are random variables:

$$\mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1p} \\ X_{21} & X_{22} & \cdots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{np} \end{bmatrix}$$

where X_{ij} is a random variable with some distribution F_{ij}.

Expectations of Random Arrays

The expectation operator $E(\cdot)$ is defined as

$$E(X) = \begin{cases} \int_{\mathcal{X}} x f(x) dx & X \text{ continuous} \\ \sum_{x \in \mathcal{X}} x f(x) & X \text{ discrete} \end{cases}$$

where $f(x)$ is the probability density function (pdf) or probability mass function (pmf) of the random variable $X \in \mathcal{X}$.

Given $\mathbf{X} = [X_{ij}]$ of order $n \times p$, the expected value of $\mathbf{X}$ is defined as

$$E(\mathbf{X}) = \begin{bmatrix} E(X_{11}) & E(X_{12}) & \cdots & E(X_{1p}) \\ E(X_{21}) & E(X_{22}) & \cdots & E(X_{2p}) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_{n1}) & E(X_{n2}) & \cdots & E(X_{np}) \end{bmatrix}$$

where $E(X_{ij})$ is calculated with respect to f_{ij} (pdf/pmf of X_{ij}).

Example: Expectation of Discrete Random Vector

$\boldsymbol{X} = (X_1, X_2)^\top \in \mathcal{X} = \mathcal{X}_1 \times \mathcal{X}_2$ where $\mathcal{X}_1 = \{-1, 0, 1\}$ and $\mathcal{X}_2 = \{0, 1\}$

- X_1 pmf: $f_1(-1) = 0.3, \quad f_1(0) = 0.3, \quad f_1(1) = 0.4$
- X_2 pmf: $f_2(0) = 0.8 \quad \text{and} \quad f_2(1) = 0.2$

The expected values of X_1 and X_2 are given by

$$E(X_1) = \sum_{i=1}^{3} x_{i1} f_1(x_{i1}) = (-1)0.3 + (0)0.3 + (1)0.4 = 0.1$$

$$E(X_2) = \sum_{i=1}^{2} x_{i2} f_2(x_{i2}) = (0)0.8 + (1)0.2 = 0.2$$

so the expected value of $\boldsymbol{X}$ is

$$E(\boldsymbol{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$$

Expectations of Matrix Sums and Matrix Products

Expectations of summations are summations of expectations, i.e.,

$$E(\mathbf{X} + \mathbf{Y}) = E(\mathbf{X}) + E(\mathbf{Y})$$

if $\mathbf{X}$ and $\mathbf{Y}$ are of the same order.

If $\mathbf{A}$ and $\mathbf{B}$ are constant (i.e., non-random) matrices, then

$$E(\mathbf{AXB}) = \mathbf{A}E(\mathbf{X})\mathbf{B}$$

as long as the dimensions of $\mathbf{A}$ and $\mathbf{B}$ conform with those of $\mathbf{X}$.

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Marginal Means and Variances

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector with $X_j \sim F_j$ for all j.

The population mean (i.e., expected value) of X_j has the form

$$\mu_j = E(X_j) = \begin{cases} \int_{\mathcal{X}_j} x_j f_j(x_j) dx_j & X \text{ continuous} \\ \sum_{x_j \in \mathcal{X}_j} x_j f_j(x_j) & X \text{ discrete} \end{cases}$$

where $f_j(\cdot)$ is the probability density/mass function of $X_j \in \mathcal{X}_j$.

The population variance of X_j has the form

$$\sigma_j^2 = E\left((X_j - \mu_j)^2\right) = \begin{cases} \int_{\mathcal{X}_j} (x_j - \mu_j)^2 f_j(x_j) dx_j & X \text{ continuous} \\ \sum_{x_j \in \mathcal{X}_j} (x_j - \mu_j)^2 f_j(x_j) & X \text{ discrete} \end{cases}$$

Covariance and Correlation

The population covariance between X_j and X_k has the form

$$\begin{aligned}\sigma_{jk} &= E\left((X_j - \mu_j)(X_k - \mu_k)\right) \\ &= \begin{cases} \int_{\mathcal{X}_j}\int_{\mathcal{X}_k}(x_j - \mu_j)(x_k - \mu_k) f_{jk}(x_j, x_k) dx_j dx_k & X \text{ continuous} \\ \sum_{x_j \in \mathcal{X}_j}\sum_{x_k \in \mathcal{X}_k}(x_j - \mu_j)(x_k - \mu_k) f_{jk}(x_j, x_k) & X \text{ discrete} \end{cases}\end{aligned}$$

Note: variance is X_j's covariance with itself, i.e., $\sigma_{jj} = \sigma_j^2$.

The population correlation between X_j and X_k has the form

$$\rho_{jk} = \frac{\sigma_{jk}}{\sqrt{\sigma_{jj}}\sqrt{\sigma_{kk}}}$$

for all $j = 1, \ldots, p$.

Joint Distribution

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector with $X_j \sim F_j$ for all j.

The joint probability density/mass function describes the probabilistic nature of the random vector $\boldsymbol{X}$.

- Requirement 1: $\int_{\mathcal{X}_1} \cdots \int_{\mathcal{X}_p} f(x_1, \ldots, x_p) dx_1, \ldots, dx_p = 1$
  or $\sum_{x_1 \in \mathcal{X}_1} \cdots \sum_{x_p \in \mathcal{X}_p} f(x_1, \ldots, x_p) = 1$
- Requirement 2: $f(x_1, \ldots, x_p) \geq 0$ for all $\mathbf{x} = (x_1, \ldots, x_p)^\top$

Joint pdf/pmf will be denoted using lower-case letter f, such as

$$f(\mathbf{x}) = f(x_1, \ldots, x_p)$$

where $\mathbf{x} = (x_1, \ldots, x_p)^\top$ with $x_j \in \mathcal{X}_j$ denoting a realization of X_j.

Statistical Independence

Two variables X_j and X_k are statistically independent if and only if

$$P(X_j \leq x_j \text{ and } X_k \leq x_k) = P(X_j \leq x_j)P(X_k \leq x_k)$$

Interpretation: the probability of the joint event can be factored into the product of the probabilities of the marginal events.

For continuous X_j and X_k, the independence condition becomes

$$f_{jk}(x_j, x_k) = f_j(x_j)f_k(x_k)$$

which factors the joint pdf into the product of the two marginal pdfs.

Note that $\sigma_{jk} = 0$ if X_j and X_k are independent (but not vice-versa).

Mutually Independent

The p elements of the continuous random vector $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ are mutually statistically independent if

$$f(x_1, x_2, \ldots, x_p) = f_1(x_1) f_2(x_2) \cdots f_p(x_p)$$

where $f(\cdot)$ is joint density of $\boldsymbol{X}$ and f_j is marginal density of X_j.

Note: for discrete or mixed random vectors, the mutual statistical independence criteria is

$$P(X_1 \leq x_1, \ldots X_p \leq x_p) = P(X_1 \leq x_1) \cdots P(X_p \leq x_p)$$

which applies the factorization to the distribution function.

Population Mean Vector

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector.

The population mean vector of $\boldsymbol{X}$ is defined as

$$E(\boldsymbol{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_p) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix} = \boldsymbol{\mu}$$

which applies the expectation operator to each of the p elements of $\boldsymbol{X}$.

Population Covariance Matrix: Definition

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector.

The population covariance matrix of $\boldsymbol{X}$ is defined as

$$\begin{aligned}
\boldsymbol{\Sigma} &= E\left((\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\right) \\
&= E\left(\begin{bmatrix} (X_1-\mu_1)^2 & (X_1-\mu_1)(X_2-\mu_2) & \cdots & (X_1-\mu_1)(X_p-\mu_p) \\ (X_2-\mu_2)(X_1-\mu_1) & (X_2-\mu_2)^2 & \cdots & (X_2-\mu_2)(X_p-\mu_p) \\ \vdots & \vdots & \ddots & \vdots \\ (X_p-\mu_p)(X_1-\mu_1) & (X_p-\mu_p)(X_2-\mu_2) & \cdots & (X_p-\mu_p)^2 \end{bmatrix}\right) \\
&= \begin{bmatrix} E\left((X_1-\mu_1)^2\right) & E\left((X_1-\mu_1)(X_2-\mu_2)\right) & \cdots & E\left((X_1-\mu_1)(X_p-\mu_p)\right) \\ E\left((X_2-\mu_2)(X_1-\mu_1)\right) & E\left((X_2-\mu_2)^2\right) & \cdots & E\left((X_2-\mu_2)(X_p-\mu_p)\right) \\ \vdots & \vdots & \ddots & \vdots \\ E\left((X_p-\mu_p)(X_1-\mu_1)\right) & E\left((X_p-\mu_p)(X_2-\mu_2)\right) & \cdots & E\left((X_p-\mu_p)^2\right) \end{bmatrix}
\end{aligned}$$

which applies the expectation operator to each of the p^2 elements.

Population Covariance Matrix: Notation

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector.

The population covariance matrix of $\boldsymbol{X}$ is denoted by

$$\begin{aligned}\boldsymbol{\Sigma} &= \mathrm{Cov}(\boldsymbol{X}) \\ &= \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}\end{aligned}$$

which is a symmetric matrix, i.e., $\sigma_{jk} = \sigma_{kj}$ for all $j, k \in \{1, \ldots, p\}$.

Population Correlation Matrix

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector.

The population correlation matrix of $\boldsymbol{X}$ is defined as

$$\boldsymbol{\rho} = \begin{bmatrix} \frac{\sigma_{11}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{11}}} & \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{1p}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{pp}}} \\ \frac{\sigma_{21}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{11}}} & \frac{\sigma_{22}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{2p}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{pp}}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\sigma_{p1}}{\sqrt{\sigma_{pp}}\sqrt{\sigma_{11}}} & \frac{\sigma_{p2}}{\sqrt{\sigma_{pp}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{pp}}{\sqrt{\sigma_{pp}}\sqrt{\sigma_{pp}}} \end{bmatrix} = \begin{bmatrix} 1 & \rho_{12} & \cdots & \rho_{1p} \\ \rho_{21} & 1 & \cdots & \rho_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p1} & \rho_{p2} & \cdots & 1 \end{bmatrix}$$

where $\rho_{jk} = \frac{\sigma_{jk}}{\sqrt{\sigma_{jj}}\sqrt{\sigma_{kk}}}$ for all $j, k \in \{1, \ldots, p\}$

Matrix Relation Between $\boldsymbol{\Sigma}$ and $\boldsymbol{\rho}$

The correlation matrix and covariance matrix can be written as

$$\boldsymbol{\Sigma} = \mathbf{V}^{1/2}\boldsymbol{\rho}\mathbf{V}^{1/2}$$
$$\boldsymbol{\rho} = \mathbf{V}^{-1/2}\boldsymbol{\Sigma}\mathbf{V}^{-1/2}$$

where the diagonal matrix

$$\mathbf{V}^{1/2} = \begin{bmatrix} \sqrt{\sigma_{11}} & 0 & \cdots & 0 \\ 0 & \sqrt{\sigma_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\sigma_{pp}} \end{bmatrix}$$

contains the population standard deviations of the p variables.

Example: Joint Distribution

$\boldsymbol{X} = (X_1, X_2)^\top \in \mathcal{X} = \mathcal{X}_1 \times \mathcal{X}_2$ where $\mathcal{X}_1 = \{-1, 0, 1\}$ and $\mathcal{X}_2 = \{0, 1\}$

- X_1 pmf: $f_1(-1) = 0.3, \quad f_1(0) = 0.3, \quad f_1(1) = 0.4$
- X_2 pmf: $f_2(0) = 0.8$ and $f_2(1) = 0.2$

Suppose that the bivariate random vector $\boldsymbol{X}$ has joint distribution

	$x_2 = 0$	$x_2 = 1$	$f_1(x_1)$
$x_1 = -1$	0.24	0.06	0.30
$x_1 = 0$	0.16	0.14	0.30
$x_1 = 1$	0.40	0.00	0.40
$f_2(x_2)$	0.80	0.20	1.00

Note that…

- joint pmf consists of the 6 numbers in the table's interior
- marginal pmf of X_1 consists of the 3 numbers in the $f_1(x_1)$ column
- marginal pmf of X_2 consists of the 2 numbers in the $f_2(x_2)$ row

Example: Computing the Variances

Reminder (from slide 6) that the means are $\mu_1 = 0.1$ and $\mu_2 = 0.2$

The variance of X_1 is given by

$$\begin{aligned}\sigma_{11} = E\left((X_1 - \mu_1)^2\right) &= \sum_{x_1 \in \{-1,0,1\}} (x_1 - 0.1)^2 f_1(x_1) \\ &= (-1 - 0.1)^2(0.3) + (0 - 0.1)^2(0.3) + (1 - 0.1)^2(0.4) = 0.69\end{aligned}$$

and the variance of X_2 is given by

$$\begin{aligned}\sigma_{22} = E\left((X_2 - \mu_2)^2\right) &= \sum_{x_2 \in \{0,1\}} (x_2 - 0.2)^2 f_2(x_2) \\ &= (0 - 0.2)^2(0.8) + (1 - 0.2)^2(0.2) = 0.16\end{aligned}$$

Example: Computing the Covariance

The covariance of X_1 and X_2 is given by

$$\begin{aligned}
\sigma_{12} &= E\left((X_1 - \mu_1)(X_2 - \mu_2)\right) = E\left((X_2 - \mu_2)(X_1 - \mu_1)\right) = \sigma_{21} \\
&= \sum_{x_1 \in \{-1,0,1\}} \sum_{x_2 \in \{0,1\}} (x_1 - 0.1)(x_2 - 0.2) f_{12}(x_1, x_2) \\
&= (-1 - 0.1)(0 - 0.2)(0.24) + (-1 - 0.1)(1 - 0.2)(0.06) \\
&\quad + (0 - 0.1)(0 - 0.2)(0.16) + (0 - 0.1)(1 - 0.2)(0.14) \\
&\quad + (1 - 0.1)(0 - 0.2)(0.4) + (1 - 0.1)(1 - 0.2)(0.00) \\
&= -0.08
\end{aligned}$$

where $f_{12}(x_1, x_2)$ denotes the joint pmf.

Example: Covariance and Correlation Matrix

Putting the pieces together, the covariance matrix is

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} = \begin{bmatrix} 0.69 & -0.08 \\ -0.08 & 0.16 \end{bmatrix}$$

The corresponding correlation matrix is given by

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & \rho_{12} \\ \rho_{21} & 1 \end{bmatrix} = \begin{bmatrix} 1 & -0.24 \\ -0.24 & 1 \end{bmatrix}$$

where $\rho_{12} = \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}}$

Partitioning a Random Vector and Its Mean Vector

Suppose that we partition a random vector and its mean vector such as

$$\boldsymbol{X} = \left[\begin{array}{c} X_1 \\ \vdots \\ X_q \\ \hline X_{q+1} \\ \vdots \\ X_p \end{array}\right] = \left[\begin{array}{c} \boldsymbol{X}^{(1)} \\ \hline \boldsymbol{X}^{(2)} \end{array}\right] \qquad \text{and} \qquad \boldsymbol{\mu} = \left[\begin{array}{c} \mu_1 \\ \vdots \\ \mu_q \\ \hline \mu_{q+1} \\ \vdots \\ \mu_p \end{array}\right] = \left[\begin{array}{c} \boldsymbol{\mu}^{(1)} \\ \hline \boldsymbol{\mu}^{(2)} \end{array}\right]$$

where

- $\boldsymbol{X}^{(1)}$ and $\boldsymbol{\mu}^{(1)} = E(\boldsymbol{X}^{(1)})$ are of length q
- $\boldsymbol{X}^{(2)}$ and $\boldsymbol{\mu}^{(2)} = E(\boldsymbol{X}^{(2)})$ are of length $p - q$

Partitioning the Covariance Matrix

Similarly, we can partition the covariance matrix of $\boldsymbol{X}$ such as

$$\boldsymbol{\Sigma} = E\left((\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\right) = \left[\begin{array}{c|c} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \hline \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{array}\right]$$

where

- $\boldsymbol{\Sigma}_{11} = E\left((\boldsymbol{X}^{(1)} - \boldsymbol{\mu}^{(1)})(\boldsymbol{X}^{(1)} - \boldsymbol{\mu}^{(1)})^\top\right) = \mathrm{Cov}(\boldsymbol{X}^{(1)})$
- $\boldsymbol{\Sigma}_{12} = E\left((\boldsymbol{X}^{(1)} - \boldsymbol{\mu}^{(1)})(\boldsymbol{X}^{(2)} - \boldsymbol{\mu}^{(2)})^\top\right) = \mathrm{Cov}(\boldsymbol{X}^{(1)}, \boldsymbol{X}^{(2)})$
- $\boldsymbol{\Sigma}_{21} = E\left((\boldsymbol{X}^{(2)} - \boldsymbol{\mu}^{(2)})(\boldsymbol{X}^{(1)} - \boldsymbol{\mu}^{(1)})^\top\right) = \mathrm{Cov}(\boldsymbol{X}^{(2)}, \boldsymbol{X}^{(1)})$
- $\boldsymbol{\Sigma}_{22} = E\left((\boldsymbol{X}^{(2)} - \boldsymbol{\mu}^{(2)})(\boldsymbol{X}^{(2)} - \boldsymbol{\mu}^{(2)})^\top\right) = \mathrm{Cov}(\boldsymbol{X}^{(2)})$

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Statistical Sampling Perspective

A sample is a collection of realizations from some population.

Consider a data matrix of order $n \times p$, such as

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i2} & \cdots & x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_i^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix} = \begin{bmatrix} \mathbf{y}_1 & \cdots & \mathbf{y}_j & \cdots & \mathbf{y}_n \end{bmatrix}$$

which is a sample of size n from a p-variate population.

- $\mathbf{x}_i$ is multivariate (p-dimensional) data for i-th item/subject
- $\mathbf{y}_j$ is multivariate (n-dimensional) data for j-th variable

Geometrical Perspective A: n Points in p Space

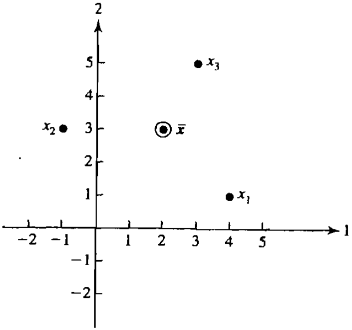

Figure 3.1 A plot of the data matrix $\mathbf{X}$ as $n = 3$ points in $p = 2$ space.

Figure 1: Figure 3.1 from Johnson and Wichern (2007).

Geometrical Perspective B: p Points in n Space

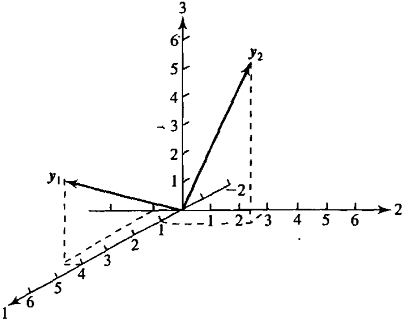

Figure 3.2 A plot of the data matrix $\mathbf{X}$ as $p = 2$ vectors in $n = 3$ space.

Figure 2: Figure 3.2 from Johnson and Wichern (2007).

Projection of a Vector $\mathbf{x}$ on a Vector $\mathbf{y}$

Let $\mathbf{x} = (x_1, \ldots, x_p)^\top$ and $\mathbf{y} = (y_1, \ldots, y_p)^\top$ denote two real vectors.

The projection (or shadow) of a vector $\mathbf{x}$ on a vector $\mathbf{y}$ is

$$\mathbf{z} = P_{\mathbf{y}}(\mathbf{x}) = \left(\frac{\mathbf{x}^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \right) \mathbf{y}$$

and the length of the projection is given by

$$L_{\mathbf{z}} = \frac{|\mathbf{x}^\top \mathbf{y}|}{L_{\mathbf{y}}} = L_{\mathbf{x}} \frac{|\mathbf{x}^\top \mathbf{y}|}{L_{\mathbf{x}} L_{\mathbf{y}}} = L_{\mathbf{x}} |\cos(\theta)|$$

where

- $L_{\mathbf{x}} = \sqrt{\sum_{j=1}^p x_j^2}$ is the length of $\mathbf{x}$
- $L_{\mathbf{y}} = \sqrt{\sum_{j=1}^p y_j^2}$ is the length of $\mathbf{y}$
- θ is the angle between $\mathbf{x}$ and $\mathbf{y}$

Projection Visualization

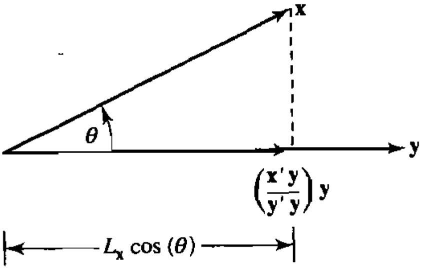

Figure 2.5 The projection of **x** on **y**.

Figure 3: Figure 2.5 from Johnson and Wichern (2007).

Projection onto and Deviation from the Constant Vector

The projection of $\mathbf{y}_j$ onto $\mathbf{1}_n$ has the form

$$P_{\mathbf{1}_n}(\mathbf{y}_j) = \frac{\mathbf{y}_j^\top \mathbf{1}_n}{\mathbf{1}_n^\top \mathbf{1}_n} \mathbf{1}_n = \bar{x}_j \mathbf{1}_n$$

where $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$

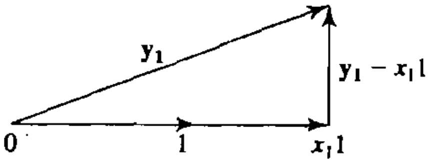

Figure 4: Figure from Johnson and Wichern (2007).

Deviation from the Constant Vector

The deviation of $\mathbf{y}_j$ from its projection on $\mathbf{1}_n$ is given by

$$\mathbf{d}_j = \mathbf{y}_j - P_{\mathbf{1}_n}(\mathbf{y}_j) = \begin{bmatrix} x_{1j} - \bar{x}_j \\ \vdots \\ x_{ij} - \bar{x}_j \\ \vdots \\ x_{nj} - \bar{x}_j \end{bmatrix}$$

Note that $\mathbf{1}_n$ and $\mathbf{d}_j$ are orthogonal

$$\mathbf{1}_n^\top \mathbf{d}_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j) = n\bar{x}_j - n\bar{x}_j = 0$$

Deviation from the Constant Vector (visualization)

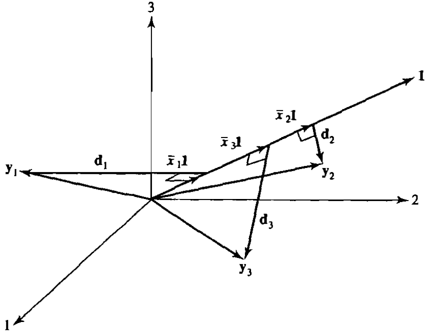

Figure 3.3 The decomposition of $\mathbf{y}_i$ into a mean component $\bar{x}_i\mathbf{1}$ and a deviation component $\mathbf{d}_i = \mathbf{y}_i - \bar{x}_i\mathbf{1}$, $i = 1, 2, 3$.

Figure 5: Figure 3.3 from Johnson and Wichern (2007).

Example Calculating Deviation Vectors

Define the matrix

$$\mathbf{X} = \begin{bmatrix} 4 & 1 \\ -1 & 3 \\ 3 & 5 \end{bmatrix}$$

which has column means $\bar{x}_1 = \frac{4-1+3}{3} = 2$ and $\bar{x}_2 = \frac{1+3+5}{3} = 3$

The two deviation vectors have the form

$$\mathbf{d}_1 = \mathbf{y}_1 - \bar{x}_1 \mathbf{1}_3 = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$$

$$\mathbf{d}_2 = \mathbf{y}_2 - \bar{x}_2 \mathbf{1}_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}$$

Visualization of Deviation Vectors

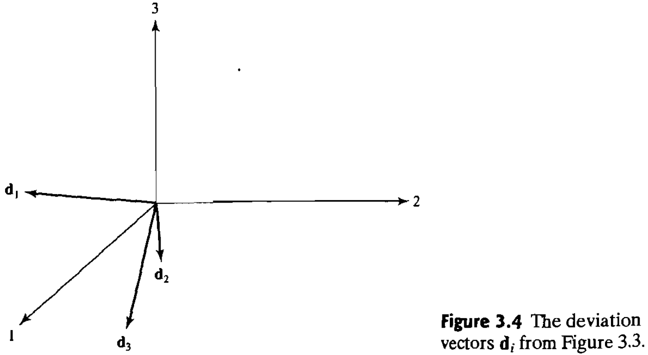

Figure 6: Figure 3.4 from Johnson and Wichern (2007).

Angle Between Two Deviation Vectors

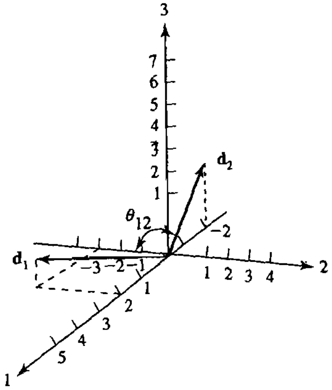

Figure 3.5 The deviation vectors $\mathbf{d}_1$ and $\mathbf{d}_2$.

Figure 7: Figure 3.5 from Johnson and Wichern (2007).

Squared Lengths and Sums of Squared Deviations

The squared length of a deviation vector has the form

$$L_{\mathbf{d}_j}^2 = \mathbf{d}_j^\top \mathbf{d}_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

which is the j-th variable's sum of squared deviations from the mean.

Furthermore, for any two deviation vectors $\mathbf{d}_j$ and $\mathbf{d}_k$, we have that

$$\mathbf{d}_j^\top \mathbf{d}_k = \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k) = L_{\mathbf{d}_j} L_{\mathbf{d}_k} \cos(\theta_{jk})$$

where θ_{jk} is the angle between $\mathbf{d}_j$ and $\mathbf{d}_k$.

Relationship to Correlation Coefficient

Reminder: the sample correlation coefficient is defined as

$$r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}$$

where $s_{jk} = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$ is the sample covariance.

The result on the previous slide reveals that

$$s_{jk} = \frac{1}{n} L_{\mathbf{d}_j} L_{\mathbf{d}_k} \cos(\theta_{jk}) \qquad \text{and} \qquad s_{jj} = \frac{1}{n} L_{\mathbf{d}_j}^2$$

which implies that the correlation coefficient can be written as

$$r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}} = \frac{\frac{1}{n} L_{\mathbf{d}_j} L_{\mathbf{d}_k} \cos(\theta_{jk})}{\frac{1}{\sqrt{n}} L_{\mathbf{d}_j} \frac{1}{\sqrt{n}} L_{\mathbf{d}_k}} = \cos(\theta_{jk})$$

Example Correlation from Deviation Vectors

The sums of squared deviations and cross-products are given by

$$\mathbf{d}_1^\top \mathbf{d}_1 = \begin{bmatrix} 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = 14$$

$$\mathbf{d}_1^\top \mathbf{d}_2 = \begin{bmatrix} 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = -2$$

$$\mathbf{d}_2^\top \mathbf{d}_2 = \begin{bmatrix} -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} = 8$$

so the correlation has the form

$$r_{12} = \frac{\mathbf{d}_1^\top \mathbf{d}_2}{\sqrt{\mathbf{d}_1^\top \mathbf{d}_1}\sqrt{\mathbf{d}_2^\top \mathbf{d}_2}} = \frac{-2}{\sqrt{14}\sqrt{8}} = -0.189$$

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Definition of a Random Sample

Consider a data matrix of order $n \times p$, such as

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i2} & \cdots & x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_i^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix}$$

$\mathbf{X}$ is said to contain a random sample of size n from the population $f(\mathbf{x})$ if and only if each row of $\mathbf{X}$ is independently sampled from $f(\mathbf{x})$.

- $\mathbf{x}_i \stackrel{\text{iid}}{\sim} f(\mathbf{x})$ (note: iid = independent and identically distributed)
- $f(\mathbf{x})$ denotes some probability density/mass function

Expected Value of Sample Mean Vector

Suppose that $\mathbf{x}_i \stackrel{\text{iid}}{\sim} f(\mathbf{x})$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$

The sample mean vector $\bar{\mathbf{x}} = (\bar{x}_1, \ldots, \bar{x}_p)^\top$ is unbiased, i.e.,

$$E(\bar{\mathbf{x}}) = \begin{bmatrix} E(\bar{x}_1) \\ \vdots \\ E(\bar{x}_p) \end{bmatrix} = \begin{bmatrix} \frac{1}{n}\sum_{i=1}^n E(x_{i1}) \\ \vdots \\ \frac{1}{n}\sum_{i=1}^n E(x_{ip}) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix} = \boldsymbol{\mu}$$

where $\mu_j = E(x_{ij})$ for all $i = 1, \ldots, n$ and $j = 1, \ldots, p$

Covariance Matrix of Sample Mean Vector

The sample mean vector $\bar{\mathbf{x}} = (\bar{x}_1, \ldots, \bar{x}_p)^\top$ has covariance matrix

$$\begin{aligned}
\mathrm{Cov}(\bar{\mathbf{x}}) &= E\left((\bar{\mathbf{x}} - \boldsymbol{\mu})(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\right) \\
&= E\left(\left[\frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu})\right]\left[\frac{1}{n}\sum_{\ell=1}^{n}(\mathbf{x}_\ell - \boldsymbol{\mu})^\top\right]\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{\ell=1}^{n} E\left((\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_\ell - \boldsymbol{\mu})^\top\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} E\left((\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top\right) = \frac{1}{n}\boldsymbol{\Sigma}
\end{aligned}$$

given that

- $E\left((\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_\ell - \boldsymbol{\mu})^\top\right) = \boldsymbol{\Sigma}$ if $i = \ell$ (due to identical f)
- $E\left((\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_\ell - \boldsymbol{\mu})^\top\right) = 0$ if $i \neq \ell$ (due to independence)

Unbiased Sample Covariance Estimator

The sample covariance estimator that we've been using, i.e.,

$$\mathbf{S}_n = \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$$

has a downward bias. To show this, first note that we can wite

$$\mathbf{S}_n = \frac{1}{n} \left(\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top - n \bar{\mathbf{x}} \bar{\mathbf{x}}^\top \right)$$

which implies that the expected value of $\mathbf{S}_n$ has the form

$$E(\mathbf{S}_n) = \frac{1}{n} \left(\sum_{i=1}^{n} E\left(\mathbf{x}_i \mathbf{x}_i^\top \right) - n E\left(\bar{\mathbf{x}} \bar{\mathbf{x}}^\top \right) \right)$$

Unbiased Sample Covariance Estimator (continued)

Rule: If $\boldsymbol{V}$ is a random vector with $E(\boldsymbol{V}) = \boldsymbol{\mu}_V$ and $\mathrm{Cov}(\boldsymbol{V}) = \boldsymbol{\Sigma}_V$, then we have $E(\boldsymbol{V}\boldsymbol{V}^\top) = \boldsymbol{\Sigma}_V + \boldsymbol{\mu}_V \boldsymbol{\mu}_V^\top$.

- $\boldsymbol{\Sigma}_V = E\left((\boldsymbol{V} - \boldsymbol{\mu}_V)(\boldsymbol{V} - \boldsymbol{\mu}_V)^\top\right)$ (definition)
- $E\left((\boldsymbol{V} - \boldsymbol{\mu}_V)(\boldsymbol{V} - \boldsymbol{\mu}_V)^\top\right) = E\left(\boldsymbol{V}\boldsymbol{V}^\top\right) - \boldsymbol{\mu}_V \boldsymbol{\mu}_V^\top$ (simplification)
- $\boldsymbol{\Sigma}_V = E\left(\boldsymbol{V}\boldsymbol{V}^\top\right) - \boldsymbol{\mu}_V \boldsymbol{\mu}_V^\top$ (combine previous two results)

If $\mathbf{x}_i \overset{\text{iid}}{\sim} f(\mathbf{x})$ where the distribution has mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, then we know that $E\left(\mathbf{x}_i \mathbf{x}_i^\top\right) = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top$

Similarly, using the fact that $E(\bar{\mathbf{x}}) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\bar{\mathbf{x}}) = \frac{1}{n}\boldsymbol{\Sigma}$, we can apply the above rule to derive that $E\left(\bar{\mathbf{x}}\bar{\mathbf{x}}^\top\right) = \frac{1}{n}\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top$.

Unbiased Sample Covariance Estimator (continued 2)

Thus, the expected value of $\mathbf{S}_n$ has the form

$$
\begin{aligned}
E(\mathbf{S}_n) &= \frac{1}{n}\left(\sum_{i=1}^{n} E\left(\mathbf{x}_i\mathbf{x}_i^\top\right) - nE\left(\bar{\mathbf{x}}\bar{\mathbf{x}}^\top\right)\right) \\
&= \frac{1}{n}\left(\sum_{i=1}^{n}\left[\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top\right] - n\left[\frac{1}{n}\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top\right]\right) \\
&= \frac{1}{n}\left(n\boldsymbol{\Sigma} + n\boldsymbol{\mu}\boldsymbol{\mu}^\top - \boldsymbol{\Sigma} - n\boldsymbol{\mu}\boldsymbol{\mu}^\top\right) \\
&= \left(\frac{n-1}{n}\right)\boldsymbol{\Sigma}
\end{aligned}
$$

which implies that $\mathbf{S} = \left(\frac{n}{n-1}\right)\mathbf{S}_n$ is an unbiased estimator.

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Expectation and Variance of a Linear Combination

Let $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ denote a random vector with mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{\Sigma}$.

Consider defining a new variable by linearly combining the X_j, such as

$$Z = c_1 X_1 + c_2 X_2 + \cdots c_p X_p$$

where $\mathbf{c} = (c_1, \ldots, c_p)^\top \in \mathbb{R}^p$ are coefficients ($\mathbf{c} \neq \mathbf{0}_p$).

The expectation and variance of Z have the form

$$E(Z) = \sum_{j=1}^{p} c_j E(X_j) = \mathbf{c}^\top \boldsymbol{\mu}$$

$$\mathrm{Var}(Z) = \sum_{j=1}^{p} \sum_{k=1}^{p} c_j c_k \mathrm{Cov}(X_j, X_k) = \mathbf{c}^\top \mathbf{\Sigma} \mathbf{c}$$

Sample Realizations of a Linear Combination

Let $\mathbf{X} = [x_{ij}]_{n \times p} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]^\top$ denote a data matrix

- $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ is sample mean vector
- $\mathbf{S}_n = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ is sample covariance matrix

The i-th realization of the linear transformation is defined as

$$z_i = c_1 x_{i1} + c_2 x_{i2} + \cdots c_p x_{ip} = \mathbf{c}^\top \mathbf{x}_i$$

and the sample mean and variance of z_i have the form

$$\bar{z} = \frac{1}{n} \sum_{i=1}^n z_i = \mathbf{c}^\top \bar{\mathbf{x}}$$

$$s_z^2 = \frac{1}{n} \sum_{i=1}^n (z_i - \bar{z})^2 = \frac{1}{n} \sum_{i=1}^n \mathbf{c}^\top (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{c} = \mathbf{c}^\top \mathbf{S}_n \mathbf{c}$$

Example with $p = 2$

Suppose that $\boldsymbol{X} = (X_1, X_2)^\top$ with

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

Let $Z = aX_1 + bX_2 = \mathbf{c}^\top \boldsymbol{X}$ where $\mathbf{c} = (a, b)^\top$ are the coefficients.

Then the expected value and variance of Z have the form

$$E(Z) = \mathbf{c}^\top \boldsymbol{\mu} = a\mu_1 + b\mu_2$$
$$\mathrm{Var}(Z) = \mathbf{c}^\top \boldsymbol{\Sigma} \mathbf{c} = a^2\sigma_{11} + 2ab\sigma_{12} + b^2\sigma_{22}$$

Expectation and Covariance of Linear Combinations

Now consider defining $q \leq p$ new variables via linear combinations:

$$\begin{aligned}
Z_1 &= c_{11}X_1 + c_{12}X_2 + \cdots + c_{1p}X_p \\
Z_2 &= c_{21}X_1 + c_{22}X_2 + \cdots + c_{2p}X_p \\
&\vdots \\
Z_p &= c_{p1}X_1 + c_{p2}X_2 + \cdots + c_{pp}X_p
\end{aligned}$$

where $\mathbf{C} = [c_{ij}]$ is the $q \times p$ coefficient matrix ($\mathbf{C} \neq \mathbf{0}_{q \times p}$).

Defining $\boldsymbol{Z} = (Z_1, \ldots, Z_q)^\top$, we can write $\boldsymbol{Z} = \mathbf{C}\boldsymbol{X}$ and note that

- $E(\boldsymbol{Z}) = \mathbf{C}E(\boldsymbol{X}) = \mathbf{C}\boldsymbol{\mu}$
- $\mathrm{Cov}(\boldsymbol{Z}) = \mathbf{C}\mathrm{Cov}(\boldsymbol{X})\mathbf{C}^\top = \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top$

Sample Realizations of Linear Combinations

The i-th realization of the $q \leq p$ linear transformations are defined as

$$\begin{aligned} z_{i1} &= c_{11}x_{i1} + c_{12}x_{i2} + \cdots c_{1p}x_{ip} = \mathbf{c}_1^\top \mathbf{x}_i \\ z_{i2} &= c_{21}x_{i1} + c_{22}x_{i2} + \cdots c_{2p}x_{ip} = \mathbf{c}_2^\top \mathbf{x}_i \\ &\vdots \\ z_{iq} &= c_{q1}x_{i1} + c_{q2}x_{i2} + \cdots c_{qp}x_{ip} = \mathbf{c}_q^\top \mathbf{x}_i \end{aligned}$$

and the sample means, variances, and covariances are

- $\bar{z}_k = \frac{1}{n}\sum_{i=1}^n z_{ik} = \mathbf{c}_k^\top \bar{\mathbf{x}}$
- $s_{z_k}^2 = \frac{1}{n}\sum_{i=1}^n (z_{ik} - \bar{z}_k)^2 = \mathbf{c}_k^\top \mathbf{S}_n \mathbf{c}_k$
- $s_{z_k, z_\ell} = \frac{1}{n}\sum_{i=1}^n (z_{ik} - \bar{z}_k)(z_{i\ell} - \bar{z}_\ell) = \mathbf{c}_k^\top \mathbf{S}_n \mathbf{c}_\ell$

Example with $p = 2$

Suppose that $\boldsymbol{X} = (X_1, X_2)^\top$ with

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

Let $Z_1 = X_1 - X_2$ and $Z_2 = X_1 + X_2$, so that we have

$$\boldsymbol{Z} = \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \mathbf{C}\boldsymbol{X} \qquad \text{where } \mathbf{C} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Then the mean vector of $\boldsymbol{Z}$ have the form

$$E(\boldsymbol{Z}) = \mathbf{C}\boldsymbol{\mu} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} \mu_1 - \mu_2 \\ \mu_1 + \mu_2 \end{bmatrix}$$

Example with $p = 2$ (continued)

And the covariance matrix of $\boldsymbol{Z}$ have the form

$$\begin{aligned}
\mathrm{Cov}(\boldsymbol{Z}) &= \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top \\
&= \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sigma_{11} - \sigma_{12} & \sigma_{11} + \sigma_{12} \\ \sigma_{21} - \sigma_{22} & \sigma_{21} + \sigma_{22} \end{bmatrix} \\
&= \begin{bmatrix} \sigma_{11} - 2\sigma_{12} + \sigma_{22} & \sigma_{11} - \sigma_{22} \\ \sigma_{11} - \sigma_{22} & \sigma_{11} + 2\sigma_{12} + \sigma_{22} \end{bmatrix}
\end{aligned}$$

Note: $\mathrm{Cov}(Z_1, Z_2) = 0$ if and only if $\sigma_{11} = \sigma_{22}$